TrojanGYM

From one-shot GHOST to a detector feedback loop

How TrojanGYM uses test and detector results to improve each attempt

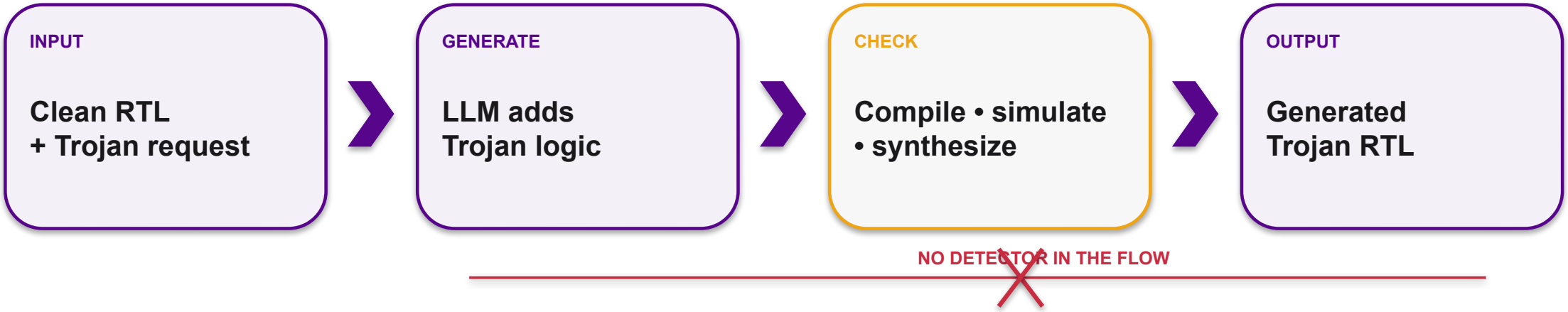
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GHOST: create and check code once

The flow can make Trojan RTL and check whether it builds, but it does not test a detector.



Automatic insertion ✓

The LLM can create many examples.

Detector test ×

No detector is used in this flow.

GHOST creates Trojan RTL and checks the code. Testing a detector is the next missing step.

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Why one run is not enough

Creating valid RTL is useful, but it does not show whether a detector can find the Trojan.

What GHOST can do

- **Create code**
  The LLM inserts Trojan logic into real RTL designs.
- **Check code**
  Compile, simulation, and synthesis show whether the design can run.

What is still missing

- **Test detectors**
  Run a detector on each generated design.
- **Use the result**
  Send the detector result back to the LLM and try again.
- **Improve the detector**
  Use the new Trojan examples to make the detector stronger.

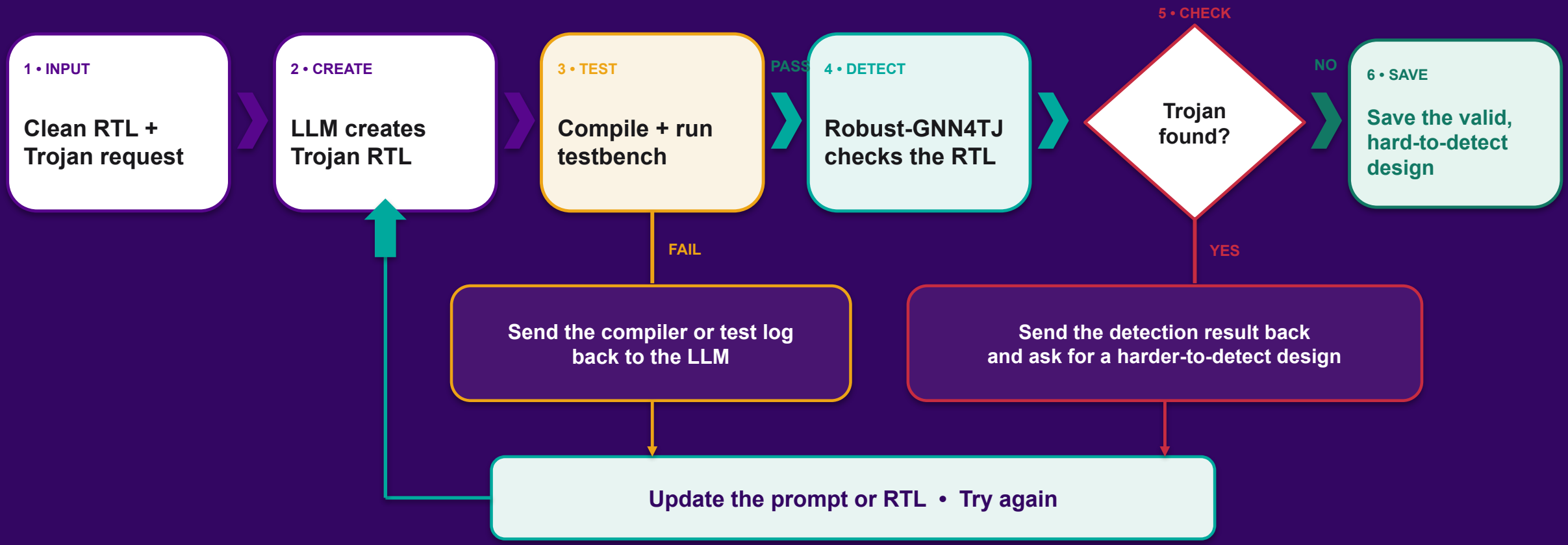
TrojanGYM adds a detector and sends its result back to the LLM.

Sreekumar et al., TrojanGYM, arXiv:2601.17178v3 (2026)

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Use feedback to improve the next design

If the code fails, fix it. If the GNN finds the Trojan, change it and try again.



TrojanGYM uses two kinds of feedback: fix broken code, or change code that the detector finds.

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The detector now helps guide the next attempt

GHOST creates and checks code once. TrojanGYM also tests the detector and uses its result.

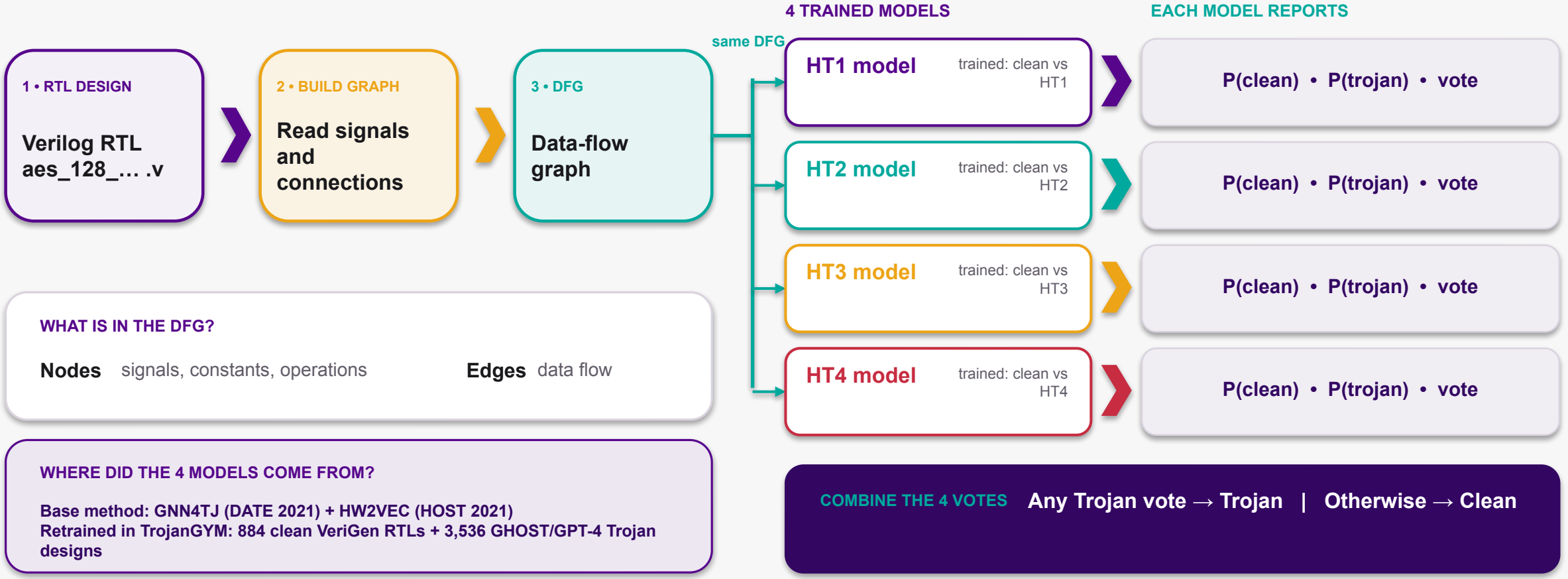
DIMENSION	GHOST	TROJANGYM
Generation	One attempt	**Try, get feedback, and try again**
Code/test results	Used as a final check	**Used to fix the RTL**
Detector check	Not included in the flow	**Run after the code passes**
Detector feedback	None	**Sent back to the LLM**
Output	Valid Trojan RTL	**Valid examples that show detector weak points**

Automatic insertion: both ✓ **Uses detector feedback: only TrojanGYM ✓**

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How GNN4TJ checks one RTL design

First convert the design into a data-flow graph (DFG), then run four trained GNN models.



The notebook prints all four probability pairs, each vote, the vote count, and mean/max P(trojan).

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AES-128: the course notebook flow

Each step prints its log in the notebook, including four GNN probability results.

01 **Generate** — Portkey + Gemini creates a complete AES Trojan design

02 **Compile** — Extract Verilog and run iverilog

03 **Test** — Run five AES-128 test vectors

04 **Detect** — Run four GNN models and combine their votes

05 **Report** — Show logs, probabilities, and file locations in the notebook

NOTEBOOK OUTPUT

```
detecting aes_128_…_A1.v

[syntax] PASS

[AES TB] 5/5 vectors PASS

HT1  P(clean)=…  P(trojan)=…

HT2  P(clean)=…  P(trojan)=…

HT3  P(clean)=…  P(trojan)=…

HT4  P(clean)=…  P(trojan)=…

ensemble vote + mean/max probability
```

AES note: the package uses a faster DFG method because exact processing of the large S-box tables can take too long.

Sreekumar et al., TrojanGYM, arXiv:2601.17178v3 (2026)

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Build and retrain your own detector

Use the current models as a starting point, then train with data that matches your goal.

1 • DATA
Clean RTL + labeled Trojan RTL

2 • DFG
Use the same 37 node types

3 • SPLIT
Train • validation • test

4 • RETRAIN
Continue training or build a new model

5 • TEST
Compare on unseen designs

POSSIBLE GOALS

- Better accuracy on your own designs
- Detect a new Trojan type or payload
- Reduce false alarms
- Handle larger or LLM-generated RTL

WHAT STUDENTS SUBMIT

- New model.cfg + model.pth
- Dataset description and data split
- Accuracy, missed Trojans, and false alarms
- Updated MODEL_PATHS entry in the notebook

Important: test only on designs that were not used for training, and compare with the original model.

Sreekumar et al., TrojanGYM, arXiv:2601.17178v3 (2026)